I ACTIVE NOISE CONTROL

Imagine living next door to Luciano Pavarotti. You love the big guy, but his full-throated practice sessions are rattling the china. To silence the booming tenor, you could do one of three things: ask him to practice his Puccini outside; wear earplugs around your apartment; or put some rock music on your CD player, crank up the volume, and drown out Pavarotti's voice. The last option wouldn't be smart, obviously, since you'd only be creating more noise. But what if the sounds coming from your CD were the acoustic mirror image of the sound waves coming out of Pavarotti's mouth? Instead of doubling the amount of noise, they would actually cancel it out, zap it from the air. What you'd be left with is peace and quiet. The idea of stifling Pavarotti or any human voice by scientific means is a bit fanciful, of course. But the theory behind it - something acoustics scientists call antinoise - is not. In fact, some of the biggest electronics companies in Japan take the concept seriously and, together with a few small American firms, are making "active noise control" (ANC) devices to help quiet the daily din. These are small but sophisticated sound systems that analyze noise digitally and then - almost instantaneously - generate equal but inverse sound waves, or antinoise. A typical ANC device can weaken a targeted noise by 10 to 15 decibels. For most industrial noise, that means reducing the sound level by 50 to 70 percent.

EXERCISE 1: Find words or phrases in the passage which mean the same as:

	COLUMN A	**COLUMN B**
a)	man *(informal word)*	
b)	make lots of short, rapid, knocking sounds	
c)	cups, saucers, plates, etc.	
d)	making a loud, deep, echoing sound	
e)	switch on or turn up (quickly) and to a very high level *(phrase)*	
f)	make one sound so loud that it is impossible to hear another *(phrase)*	
g)	clever	
h)	reflection which is either exactly the same, or the same but reversed *(phrase)*	
i)	eliminate by "shooting"	
j)	stop from continuing; suppress	
k)	imaginary; not based on reality	
l)	long-lasting, loud and unpleasant noise	
m)	immediately; without delay	
n)	reversed; exactly opposite	

EXERCISE 2: Choose the correct answer according to the passage.

1. According to the passage, the new device . . . . . .

A) can make the noise it's aimed at 10 to 15 decibels quieter
B) lowers all industrial noise by at least half
C) generates sound waves of 10 to 15 decibels
D) can weaken a noise of 10 to 15 decibels by 50 to 70 per cent
E) instantaneously reduces noise to a level of 10 to 15 decibels

2. According to the writer, it is unrealistic to . . . . . .

A) wear earplugs to block out noise
B) expect to live in peace and quiet
C) take the theory of antinoise seriously
D) suppress human voices scientifically
E) play rock music CDs too loudly

3. The device manufactured by some Japanese and American firms . . . . . .

A) has solved the problem of industrial noise
B) generates sound waves which are much louder than the targeted noise
C) produces sound waves which partially cancel out the problem noise
D) is not big enough to be able to eliminate all noise
E) reflects the problem sound waves back onto themselves

EXERCISE 3: Complete the sentences by selecting words from Column B in EXERCISE 1.

1. That child is obviously very……because she can already beat just about anybody at cards or video games, and she's only seven.
2. With so much traffic on the roads it is……to think you can drive to work in just ten minutes.
3. We must get that window fixed. Everytime the wind blows, you can hear it ……in its frame.
4. Being able to understand words and sentences……, without having to translate them first, is a sign of good progress when learning a language.
5. We were unable to continue with the lesson because the noise from the building work ……the teacher's voice.